

Classical Cryptography and Perfect Secrecy

CS 171: Cryptography

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Overview

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- 3 Design Principles
- 4 Defining Secure Encryption
- 5 Attempts at Defining Security
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Recap from Previous Lecture

- **Private-key encryption** (Gen, Enc, Dec): Alice sends $c \leftarrow \text{Enc}_k(m)$, and Bob recovers $m := \text{Dec}_k(c)$ using the shared secret key k .

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- The **shift cipher**: shift every letter by $k \in \{0, \dots, 25\}$ positions (mod 26) — insecure, since a **brute-force** attack tries all 26 keys.

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- Today: more classical ciphers and their lessons, then the principles of modern cryptography and our first security definition.

From an Art to a Science

Last time

Cryptography has developed from an **art** to a **science**.

Is the Shift Cipher Secure?

The key was $k = 2$ (the letter 'c')

ciphertext: etarvqogcpuetarvqitcrja

key: cccccccccccccccccccccccc

plaintext: cryptomeanscryptography

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Exhaustive search over k

c : etarvqogcpuetarvqitcrja
 $k = 1$: dszqunfnbotdszquphsbqiz
 $k = 2$: cryptomeanscryptography (✓ makes sense)
 $k = 3$: bqzosnldzmrqbzosnfqzogx
⋮ ⋮

Lesson: Large Key Space

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Any secure encryption scheme must have a sufficiently **large key space** (so as to make a brute-force attack computationally infeasible).

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Any secure encryption scheme must have a sufficiently **large key space** (so as to make a brute-force attack computationally infeasible).

- **Not sufficient:** the next cipher has a huge key space, yet is easily broken.
- Also, a brute-force attack identifies the key only once the ciphertext is long enough (more details later).

The Mono-alphabetic Substitution Cipher

- $\mathcal{K} = \{\text{all bijections (or permutations) from } \{a, \dots, z\} \text{ to } \{a, \dots, z\}\}.$
- An example permutation/key π would be:

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- $\mathcal{K} = \{\text{all bijections (or permutations) from } \{a, \dots, z\} \text{ to } \{a, \dots, z\}\}.$
- An example permutation/key π would be:
- How large is the key space now?

$$26! \approx 2^{88}$$

CIPHER ALPHABET

A = B	H = A	O = O	V = L
B = V	I = D	P = Y	W = P
C = G	J = Z	Q = F	X = U
D = Q	K = C	R = J	Y = I
E = K	L = W	S = X	Z = R
F = M	M = S	T = H	
G = N	N = E	U = T	

The Mono-alphabetic Substitution Cipher, Formally

Definition 1 (Mono-alphabetic Substitution Cipher)

- Gen: choose uniform $k = \pi \in \mathcal{K}$.
- $\text{Enc}_k(m_1 \cdots m_t)$: output $c_1 \cdots c_t$, where $c_i := \pi(m_i)$.
- $\text{Dec}_k(c_1 \cdots c_t)$: output $m_1 \cdots m_t$, where $m_i := \pi^{-1}(c_i)$.

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Correctness

For each i : $\pi^{-1}(c_i) = \pi^{-1}(\pi(m_i)) = m_i$.

Is the Mono-alphabetic Substitution Cipher Secure?

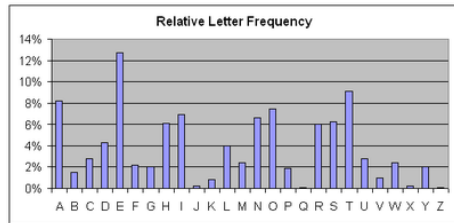
Is the Mono-alphabetic Substitution Cipher Secure?

No — frequency analysis

The substitution is fixed, so the frequency of each ciphertext letter equals the frequency of the corresponding plaintext letter. Match observed frequencies against known English frequencies.

- Sample ciphertext:

jgrmqoyghm vbjwrwqfpwhgf...



Relative letter frequency in English.

Lesson: Smooth the Frequency Distribution

Lesson

Need to “smooth out” the frequency distribution of characters
(rule out statistical tests).

- This needs a larger ciphertext than was needed to break the shift cipher.

Vigenère Cipher (Multiple Shift Cipher)

$\mathcal{K} = \{a, \dots, z\}^t$ for a period t (description of the scheme by example)

Example (from Katz and Lindell)

plaintext:	tellhimaboutme
key:	<u>cafecafecafeca</u>
ciphertext:	veq pjiredozxoe

- How large is the key space? 26^t ($\approx 2^{70}$ for $t = 15$).

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- **Smooth out the distribution:** the two 'l's in 'tell' are encrypted differently.
- Remained unbroken for centuries.

Is the Vigenère Cipher Secure?

Attack (when t is known)

- For each $j \in \{1, \dots, t\}$, consider the stream $c_j, c_{j+t}, c_{j+2t}, \dots$
 - Every letter in this stream is shifted by the *same* value k_j — it is a shift-cipher encryption.
 - Can do frequency analysis: guess that its most frequent letter corresponds to 'e' and recover k_j .
-
- Next: a more methodical attack (removing the guesswork).

Better Attack on the Shift Cipher

- Let p_i ($0 \leq i \leq 25$) denote the frequency of the i th English letter in normal English plaintext.
 - One can compute $\sum_i p_i^2 \approx 0.065$.
- Let q_i ($0 \leq i \leq 25$) denote the frequency of the i th English letter in the given ciphertext.
 - If the key is k , then $q_{i+k} \approx p_i$ for every i .

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- Output the $j \in \{0, \dots, 25\}$ for which $I_j = \sum_i p_i q_{i+j}$ (subscripts mod 26) is closest to 0.065: this j is (very likely) the key.
 - $I_k \approx \sum_i p_i^2 \approx 0.065$, while a wrong shift misaligns the frequencies, giving $I_j \approx 0.038$.

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Vigenère cipher with known period t : apply the same attack separately to each stream $c_\beta, c_{\beta+t}, c_{\beta+2t}, \dots$ for $\beta \in \{1, \dots, t\}$.

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- **Simple strategy:** try all possible choices of t !
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- **More efficient**: for a given τ , let r_i ($0 \leq i \leq 25$) denote the frequency of the i th English letter in the stream $c_1, c_{1+\tau}, c_{1+2\tau}, \dots$ and compute

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- If $\tau = t$: the stream is a single shift cipher, so $S_\tau \approx \sum_i p_i^2 \approx 0.065$.
- Otherwise: the stream mixes letters encrypted under different shifts, so $r_i \approx \frac{1}{26}$ and $S_\tau \approx 26 \times \frac{1}{26^2} = 0.038$ (heuristically).
- Output the smallest τ with $S_\tau \approx 0.065$ (multiples of t pass the test too).

Lesson: Ad Hoc Fixes Break

Lesson

Ad hoc fixes are likely to break.

Three pillars of modern cryptography

- ① **Definitions** — formally specify what security means.
 - ② **Assumptions** — state precisely what is assumed to be hard.
 - ③ **Proofs of security** — show a construction meets the definition under the assumptions.
- We study each in turn, then use them to define **secure encryption**.

Importance of Definitions

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Definitions also

Communicate security properties — useful when the scheme is used inside a larger system.

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- We need assumptions — e.g. $P \neq NP$, and in fact more!
- An assumption should be **clearly stated**.
 - We can then validate / invalidate it.
 - We can compare schemes based on different assumptions.

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A construction satisfies the considered security definition under the specified assumptions.

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Limitations

- The **definition** may not capture the real-world attack space.
- The **assumption** might be invalid.
- The **implementation** might be buggy.

Crypto Is Still a Bit of an Art

- Crypto remains a bit of an **art**!
- Yet, definitions and security proofs are immensely valuable — they **reduce the attack space**.

Defining Secure Encryption

Goal

Define **perfect** secrecy.

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- This setting illustrates **definitions** and **proofs**...
- ...but **not assumptions** for now (perfect secrecy needs none).

Private-Key Encryption (Syntax)

Definition 2 (Private-Key Encryption Scheme)

A **private-key encryption scheme** is a message space $\mathcal{M}$, a key space $\mathcal{K}$, and algorithms (Gen, Enc, Dec):

- Gen (key generation): outputs $k \in \mathcal{K}$.
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For all $m \in \mathcal{M}$ and k output by Gen:
 $\text{Dec}_k(\text{Enc}_k(m)) = m$.

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Secret

k must be kept secret.

Attempt at Security?

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This is the right idea — let us make it precise.

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Informal definition

Regardless of any information an attacker already has, a ciphertext should leak **no additional information** about the plaintext.

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Caveat

What about the **length** of the message? The ciphertext may reveal the message length; perfect secrecy is over a *fixed-length* message space.

Perfect Security: Formally

No assumptions!

Perfect secrecy is an **information-theoretic** notion: it holds against a computationally *unbounded* adversary and rests on **no computational assumptions**.

- We first fix notation, then give three equivalent definitions.

Notation

Fix $\mathcal{M}$, $\mathcal{K}$ and $(\text{Gen}, \text{Enc}, \text{Dec})$. Treat the key, message, and ciphertext as **random variables**.

Random variable

A variable that takes on (discrete) values with certain probabilities.

- K : random variable for the output of $\text{Gen}()$, so $\Pr[K = k] = \Pr[\text{Gen}() \text{ outputs } k]$.
- M : random variable for the message, ranging over $\mathcal{M}$. If $\Pr[M = m] > 0$ then $m \in \mathcal{M}$. (*Application specific, e.g.* $\Pr[M = \text{"Attack!"}] = .6$, $\Pr[M = \text{"Retreat"}] = .4$.)
- C : random variable for the ciphertext, $\Pr[C = c] = \Pr[\text{Enc}_K(M) = c]$ (over the randomness of Enc as well).

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- C : random variable for the ciphertext, $\Pr[C = c] = \Pr[\text{Enc}_K(M) = c]$ (over the randomness of Enc as well).

Key independence

M and K are **independent**!

Shift Cipher: Example 1

Setup

Shift cipher, $k \in \{0, \dots, 25\}$, single-character messages with $\Pr[M = \text{'a'}] = 0.6$, $\Pr[M = \text{'b'}] = 0.4$.

- What is $\Pr[K = k]$?

Shift Cipher: Example 1

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- What is $\Pr[K = k]$? $= 1/26$.
- What is $\Pr[C = 'z']$?

$$\begin{aligned}\Pr[C = 'z'] &= \Pr[\text{Enc}_K(M) = 'z'] \\ &= .6 \times \Pr[\text{Enc}_K('a') = 'z'] + .4 \times \Pr[\text{Enc}_K('b') = 'z'] \\ &= .6 \times \frac{1}{26} + .4 \times \frac{1}{26} = \frac{1}{26}.\end{aligned}$$

Shift Cipher: Example 2

Setup

Two-character messages: $\Pr[M = \text{'aa'}] = 0.6$, $\Pr[M = \text{'ab'}] = 0.4$. Observe $C = \text{'zz'}$.

- Can you guess what m is?

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- Only $m = \text{'aa'}$ can encrypt to 'zz' (both characters shift by the same k), so $m = \text{'aa'}$ with certainty.

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- Can you guess what m is?
- Only $m = \text{'aa'}$ can encrypt to 'zz' (both characters shift by the same k), so $m = \text{'aa'}$ with certainty.

Conclusion

The shift cipher is **insecure** even for messages of length two.

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Definition 3 (Definition 1 — Perfect Secrecy)

An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is **perfectly secret** if for every probability distribution over $\mathcal{M}$, every message $m \in \mathcal{M}$, and every ciphertext c with $\Pr[C = c] > 0$:

$$\Pr[M = m \mid C = c] = \Pr[M = m].$$

Example

Setup

Shift cipher, $k \in \{0, \dots, 25\}$, $\Pr[M = 'a'] = 0.6$, $\Pr[M = 'b'] = 0.4$.

We have $\Pr[K = k] = 1/26$ and $\Pr[C = 'z'] = 1/26$. By Bayes:

$$\begin{aligned}\Pr[M = 'a' \mid C = 'z'] &= \frac{\Pr[C = 'z' \mid M = 'a'] \cdot \Pr[M = 'a']}{\Pr[C = 'z']} \\ &= \frac{\frac{1}{26} \cdot 0.6}{\frac{1}{26}} = 0.6 = \Pr[M = 'a'].\end{aligned}$$

Bayes' Theorem

$$\Pr[A \mid B] = \frac{\Pr[B \mid A] \cdot \Pr[A]}{\Pr[B]}.$$

Definition 2

Definition 4 (Definition 2 — Perfect Secrecy)

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where the probability is only over K and the random coins of Enc .

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Claim

Definition 1 is equivalent to Definition 2.

Definition 2 $\Rightarrow$ Definition 1

Given / To prove

Given: $\forall m, m', c : \Pr[\text{Enc}_K(m) = c] = \Pr[\text{Enc}_K(m') = c]$.

To prove: for every distribution on $\mathcal{M}$, every m , and every c with $\Pr[C = c] > 0$:
 $\Pr[M = m \mid C = c] = \Pr[M = m]$.

By Bayes' theorem and the law of total probability,

$$\begin{aligned}\Pr[M = m \mid C = c] &= \frac{\Pr[C = c \mid M = m] \cdot \Pr[M = m]}{\Pr[C = c]} = \frac{\Pr[\text{Enc}_K(m) = c] \cdot \Pr[M = m]}{\sum_{m'} \Pr[C = c \mid M = m'] \cdot \Pr[M = m']} \\ &= \frac{\Pr[\text{Enc}_K(m) = c] \cdot \Pr[M = m]}{\sum_{m'} \Pr[\text{Enc}_K(m) = c] \cdot \Pr[M = m']} = \frac{\Pr[M = m]}{\sum_{m'} \Pr[M = m']} = \Pr[M = m].\end{aligned}$$

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Try on your own

Definition 1 $\Rightarrow$ Definition 2.

Definition 3 (Game Style)

Game $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}$

(eav = eavesdropper)

- ① $\mathcal{A}$ outputs $m_0, m_1 \in \mathcal{M}$.
- ② Challenger samples $b \leftarrow \{0, 1\}$, $k \leftarrow \text{Gen}()$, and the **challenge ciphertext** $c \leftarrow \text{Enc}_k(m_b)$ is given to $\mathcal{A}$.
- ③ $\mathcal{A}$ outputs b' .
- ④ **Output** 1 if $b = b'$, and 0 otherwise.

Definition 3 (Game Style)

Game $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}$

(eav = eavesdropper)

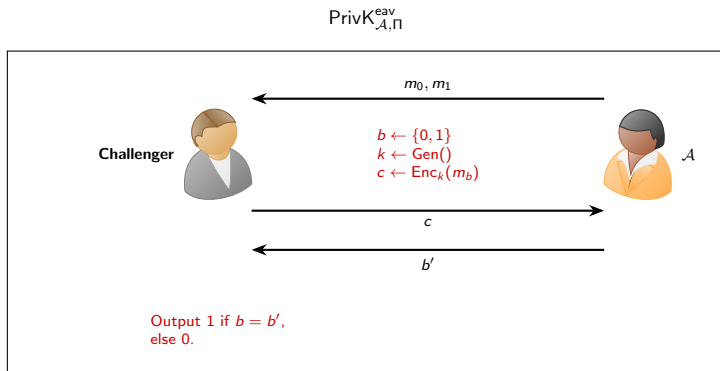
- ① $\mathcal{A}$ outputs $m_0, m_1 \in \mathcal{M}$.
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Definition 3 — Perfect Indistinguishability

$\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is **perfectly indistinguishable** if for all (even unbounded) $\mathcal{A}$:

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] = \frac{1}{2}.$$

$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}$: Challenger vs Adversary



Remark

$\mathcal{A}$ can always succeed with probability $\frac{1}{2}$. How? Just guess b' at random — so $\frac{1}{2}$ is the best achievable, hence the definition.

Equivalence of the Definitions

Lemma 5

*An encryption scheme is **perfectly secret** if and only if it is **perfectly indistinguishable**.*

- Together with the earlier equivalence, all three definitions coincide.
- **Prove on your own.**

Summary

- Classical ciphers and how they break:
 - Shift cipher — tiny key space (brute force).
 - Mono-alphabetic substitution — frequency analysis.
 - Vigenère — reduces to several shift ciphers once the period is found.
- **Lessons:** large key space (necessary), smooth statistics, and ad hoc fixes break.
- Modern crypto rests on three pillars: **definitions, assumptions, proofs.**
- **Perfect secrecy** is information-theoretic — three equivalent definitions:
 - **Def. 1:** $\Pr[M = m \mid C = c] = \Pr[M = m]$.
 - **Def. 2:** $\Pr[\text{Enc}_K(m) = c] = \Pr[\text{Enc}_K(m') = c]$.
 - **Def. 3:** perfect indistinguishability, $\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] = \frac{1}{2}$.

Reading

Today: KL 1.3–1.4, 2.1. Next time: perfect secrecy continued, the one-time pad, and computational security (KL 2.2–2.3, 3.1).